Lecture 7:
Set Operations

Section 2.2

Section Summary

- Set Operations
  - Union
  - Intersection
  - Complementation
  - Difference
- Set Cardinality: Inclusion, Exclusion
- Set Identities
- Proving Identities

Union

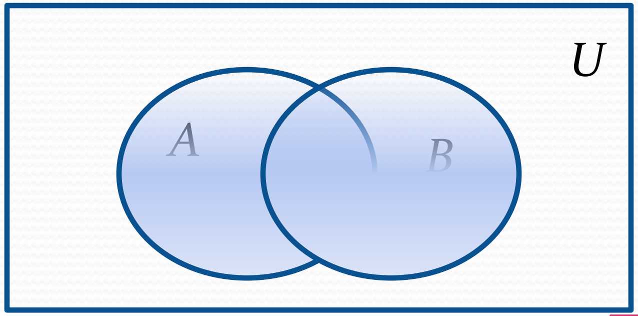
- **Definition** : Let A and B be sets. The *union* of the sets A and B, denoted by $A \cup B$, is the set:

$$\{x | x \in A \vee x \in B\}$$

- **Example** : What is $\{1,2,3\} \cup \{3, 4, 5\}$?

Solution : $\{1,2,3,4,5\}$

Venn Diagram for $A \cup B$



Intersection

- **Definition** : The *intersection* of sets A and B, denoted by $A \cap B$, is

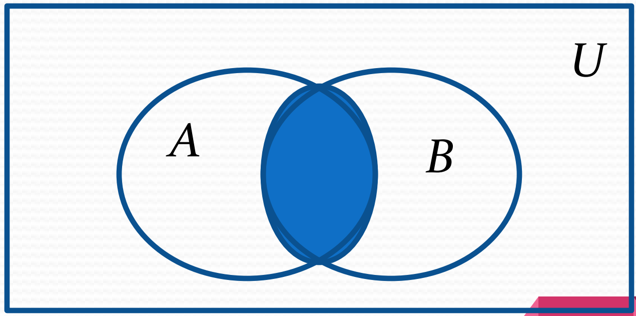
$$\{x | x \in A \wedge x \in B\}$$

- Note if the intersection is empty, then A and B are said to be *disjoint*.
- **Example** : What is $\{1,2,3\} \cap \{3,4,5\}$?

  Solution : $\{3\}$
- **Example:** What is $\{1,2,3\} \cap \{4,5,6\}$?

  Solution : $\emptyset$

Venn Diagram for $A \cap B$



Complement

Definition : If A is a set, then the complement of the A (with respect to U), denoted by $\bar{A}$ is the set $U - A$.

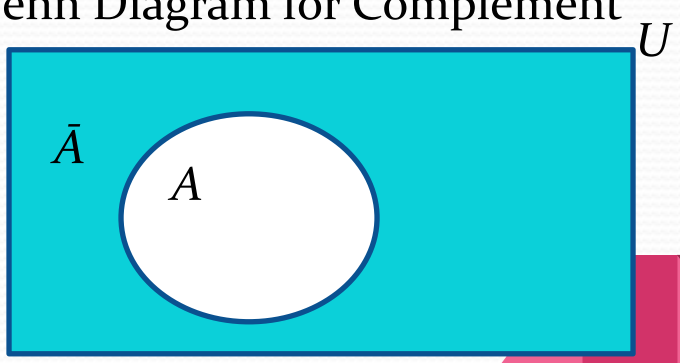
$$\bar{A} = \{x \in U \mid x \notin A\}$$

(The complement of A is sometimes denoted by A^c.)

Example : If U is the positive integers less than 100, what is the complement of $\{x \mid x > 70\}$?

Solution: $\{x \mid x \leq 70\}$

Venn Diagram for Complement

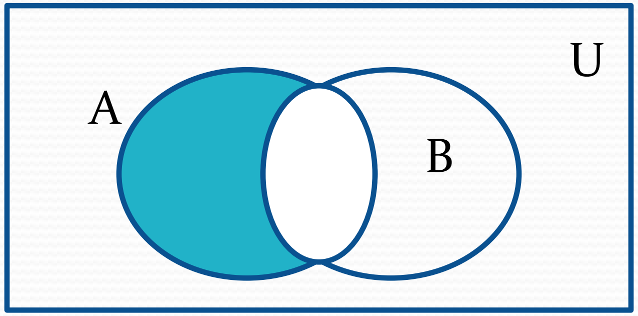


Difference

- **Definition** : Let A and B be sets. The *difference* of A and B, denoted by $A - B$, is the set containing the elements of A that are not in B. The difference of A and B is also called the complement of B with respect to A.

$$A - B = \{x \mid x \in A \land x \notin B\} = A \cap B^c$$

Venn Diagram for $A - B$



Review Questions

Example : $U = \{0,1,2,3,4,5,6,7,8,9,10\}$ $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7,8\}$

a. $A \cup B$

Solution: $\{1,2,3,4,5,6,7,8\}$

b. $A \cap B$

Solution: $\{4,5\}$

c. $\bar{A}$

Solution: $\{0,6,7,8,9,10\}$

d. $\bar{B}$

Solution: $\{0,1,2,3,9,10\}$

e. $A - B$

Solution: $\{1,2,3\}$

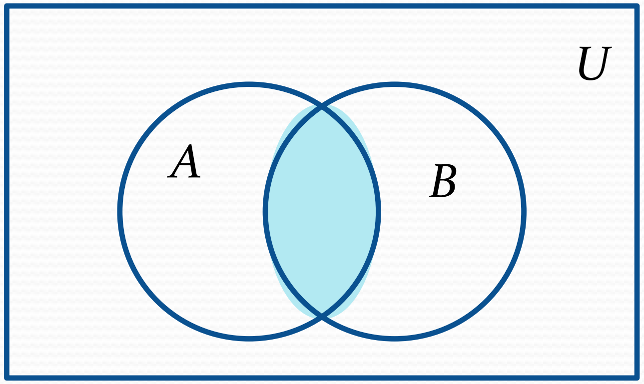
f. $B - A$

Solution: $\{6,7,8\}$

The Cardinality of the Union of Two Sets

- Inclusion-Exclusion Principle

$$|A \cup B| = |A| + |B| - |A \cap B|$$



- **Example** : Let A be the math majors in your class and B be the CS majors. To count the number of students who are either math majors or CS majors:
  - Add the number of math majors and the number of CS majors.
  - Then subtract the number of joint CS/math majors.

Inclusion-Exclusion Principle

Example: A restaurant served 50 guests one day. At the end, there was 37 orders of burgers and 29 orders of smoothies. If everyone ordered a burger or a smoothie that day, how many people ordered both?

Solution:

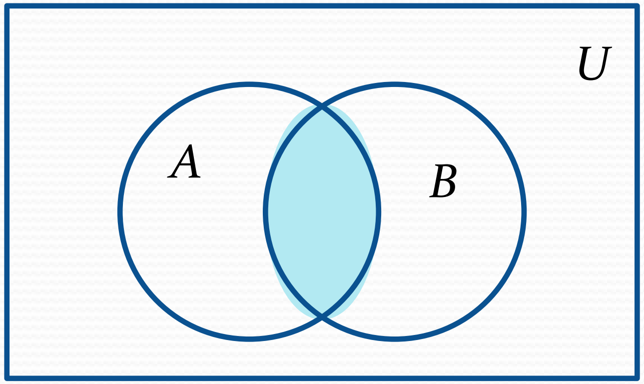
Let A and B be the sets of people who ordered burgers and smoothies respectively. Then, $|A| = 37$ and $|B| = 29$.

Since, $|A \cup B| = 50$

$=> \ |A| + |B| - |A \cap B| = 50$

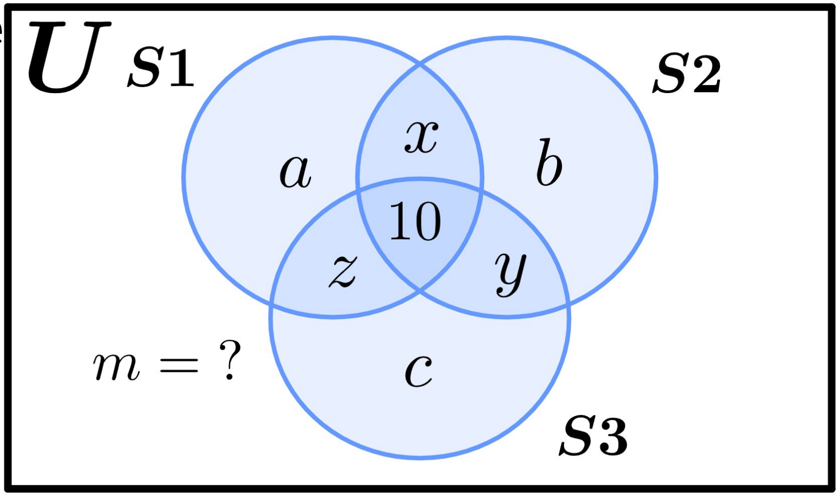
$=> |A \cap B| = |A| + |B| - 50$

$=> |A \cap B| = 37 + 29 - 50 = 16.$



Inclusion-Exclusion Principle

Example: A coding competition had 100 students participating. 50 of the participants were able to solve the first problem. However, for the second and third problem, 39 and 23 students were able to solve them, respectively. **If 10 students solved all three problems and 12 solved any 2, how many students solved none**



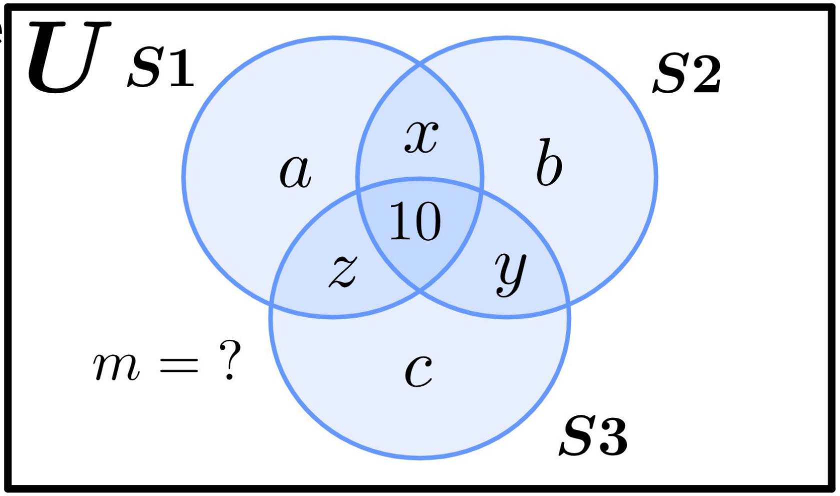
Solution: Let, S1, S2 and S3 be the sets of students who answered the first, second and third problem respectively, & U be the universal set of all participants.

Then, $|S1| = 50 = a+x+z+10$,
$|S2| = 39 = b+x+y+10$, $|S3| = 23 = c+y+z+10$ and $|U| = 100$.
Inferring from the Venn Diagram,

Number of students that solved zero problems = m,
Number of students that solved 1 problem = a+b+c,
Number of students that solved 2 problems = x+y+z = 12 (given),
Number of students that solved all 3 problems = 10.

Inclusion-Exclusion Principle

Example: A coding competition had 100 students participating. 50 of the participants were able to solve the first problem. However, for the second and third problem, 39 and 23 students were able to solve them, respectively. **If 10 students solved all three problems and 12 solved any 2, how many students solved none**



Solution (continued):

Here, |S1|+|S2|+|S3| = 50+39+23

=> (a+x+z+10)+(b+x+y+10)+ (c+y+z+10) = 112

=> a+b+c+2x+2y+2z = 82

=> a+b+c = 82 – 2(x+y+z)

=> a+b+c = 82 – 2*12 = 58

So, 58 students solved exactly 1 problem.

Also, |U| = m+(a+b+c)+(x+y+z)+10 = 100.

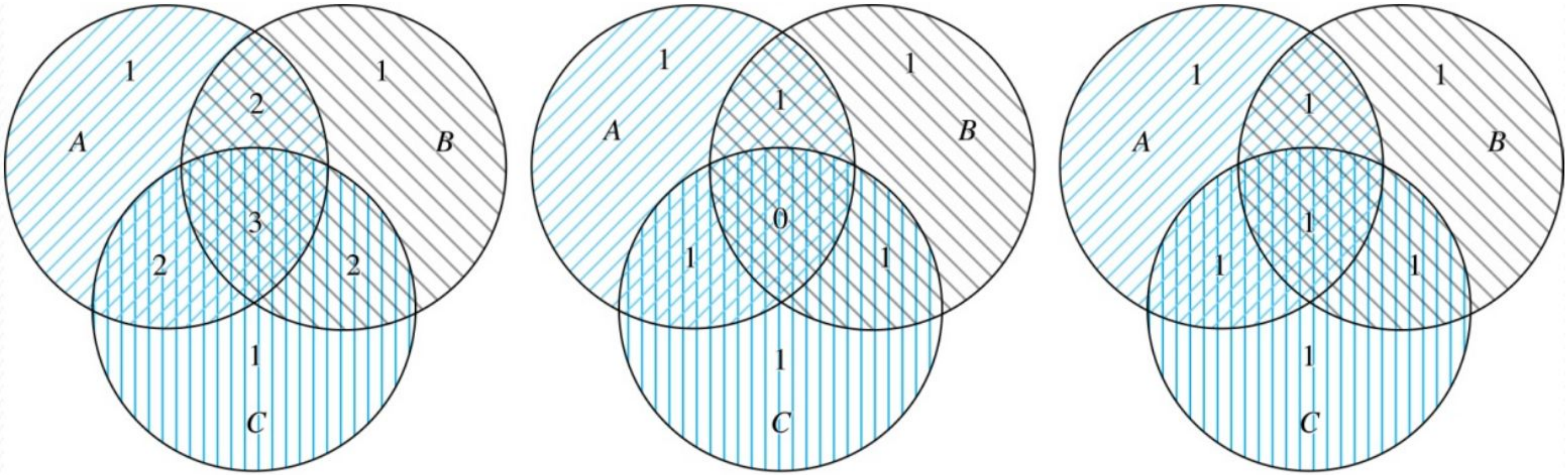
=> m+58+12+10=100 => m = 100–80 = 20.

Thus, 20 students solved zero problems.

Inclusion-Exclusion Principle for 3 Sets

- For 3 sets, the Inclusion-Exclusion Principle becomes:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$



(a) Count of elements by $|A|+|B|+|C|$

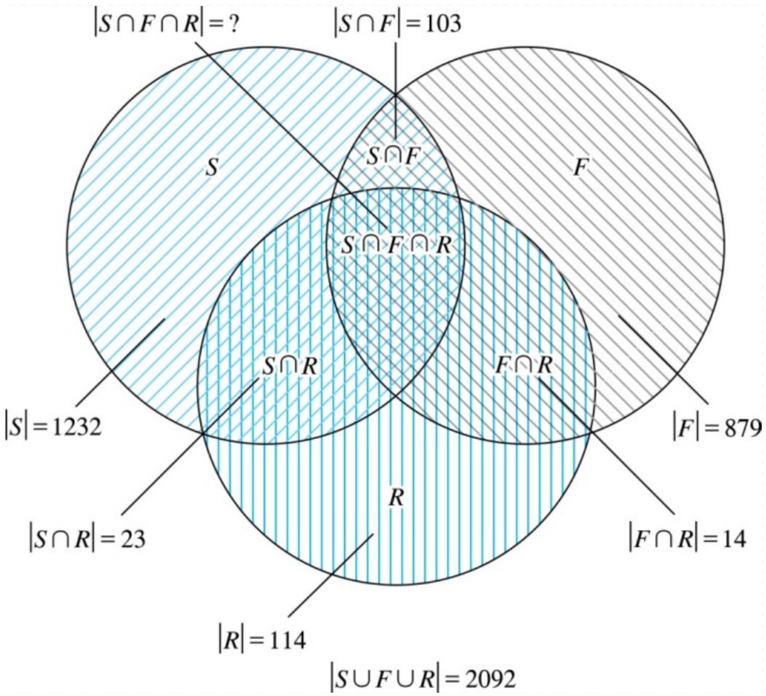
(b) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|$

(c) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|+|A\cap B\cap C|$

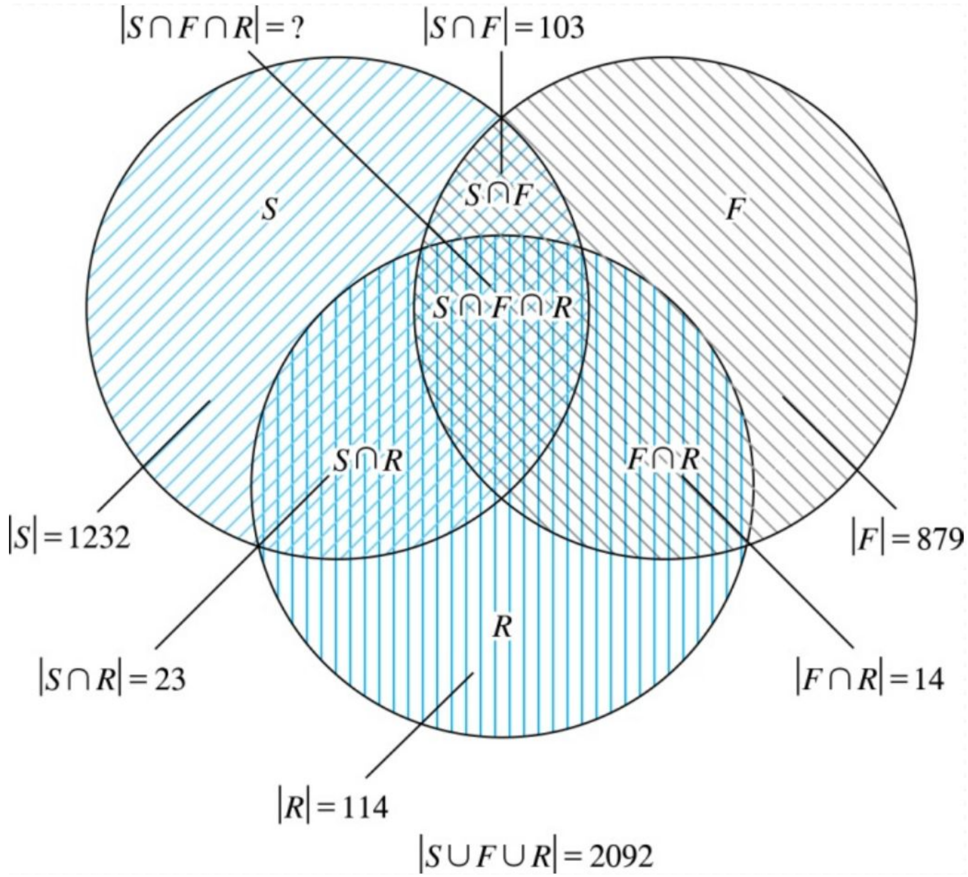
Inclusion-Exclusion Principle for 3 Sets

- **Example:** In a class,

1232 students have taken a course in Spanish,
879 have taken a course in French, and
114 have taken a course in Russian. Further,
103 have taken courses in both
Spanish and French,
23 have taken courses in both
Spanish and Russian, and
14 have taken courses in both
French and Russian.
If 2092 students have taken a course in
at least one of Spanish, French and Russian,
how many students have taken a course in all 3 languages?



Inclusion-Exclusion Principle for 3 Sets



- **Example (Cont.):**

Here, $|S∪F∪R| = |S|+ |F|+ |R| - |S∩F| - |S∩R| - |F∩R| + |S∩F∩R|$

=> $2092 = 1232 + 879 + 114 -103 -23 -14 + |S∩F∩R|$.

=> $|S∩F∩R| = 7$.

Set Identities

TABLE 1 **Set Identities.**

Identity	*Name*
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

Proving Set Identities

- Different ways to prove set identities:
  1. Prove that **each set (side of the identity) is a subset of the other** .
  2. Use set builder notation and **propositional logic** .
  3. **Membership Tables:** Verify that elements in the same combination of sets always either belong or do not belong to the same side of the identity. Use 1 to indicate it is in the set and a 0 to indicate that it is not.

Proving Set Identities: Second De Morgan Law

Example : Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Solution : We prove this identity by showing that:

1) $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$ and

2) $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

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Proving Set Identities: Second De Morgan Law (Cont.)

1) Show that: $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$

$x \in \overline{A \cap B}$	by assumption
$x \notin A \cap B$	defn. of complement
$\neg((x \in A) \wedge (x \in B))$	defn. of intersection
$\neg(x \in A) \vee \neg(x \in B)$	1st De Morgan Law for Prop Logic
$x \notin A \vee x \notin B$	defn. of negation
$x \in \overline{A} \vee x \in \overline{B}$	defn. of complement
$x \in \overline{A} \cup \overline{B}$	defn. of union

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Proving Set Identities: Second De Morgan Law (Cont.)

2) Show that: $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

$x \in \overline{A} \cup \overline{B}$	by assumption
$(x \in \overline{A}) \vee (x \in \overline{B})$	defn. of union
$(x \notin A) \vee (x \notin B)$	defn. of complement
$\neg(x \in A) \vee \neg(x \in B)$	defn. of negation
$\neg((x \in A) \wedge (x \in B))$	by 1st De Morgan Law for Prop Logic
$\neg(x \in A \cap B)$	defn. of intersection
$x \in \overline{A \cap B}$	defn. of complement

Proving Set Identities with Propositional Logic

We can also prove the second De Morgan's law by the following process with propositional logic on Set Builder Notation:

$$
\begin{aligned}
\overline{A \cap B} &= \{x | x \notin A \cap B\} && \text{by defn. of complement} \\
&= \{x | \neg(x \in (A \cap B))\} && \text{by defn. of does not belong symbol} \\
&= \{x | \neg(x \in A \wedge x \in B\} && \text{by defn. of intersection} \\
&= \boxed{\{x | \neg(x \in A) \vee \neg(x \in B)\} \quad \text{by 1st De Morgan law for Prop Logic}} \\
&= \{x | x \notin A \vee x \notin B\} && \text{by defn. of not belong symbol} \\
&= \{x | x \in \overline{A} \vee x \in \overline{B}\} && \text{by defn. of complement} \\
&= \{x | x \in \overline{A} \cup \overline{B}\} && \text{by defn. of union} \\
&= \overline{A} \cup \overline{B} && \text{by meaning of notation}
\end{aligned}
$$

Therefore, $\overline{A \cap B} = \overline{A} \cup \overline{B}$. ◀

Proving Set Identities with Membership Tables

We can use a membership table to show that **$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$** :

TABLE 2 A Membership Table for the Distributive Property.

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0